Light

GCSE Computer Science - Cyber Security Threats (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What three aspects of data and systems does cyber security aim to protect?

- A. Speed, colour, and size
- B. Confidentiality, integrity, and availability
- C. Price, location, and weight
- D. Temperature, humidity, and pressure

Q2 (1 mark)

Why can social engineering attacks succeed even against organisations with strong technical security?

- ○ **A.** Because social engineering attacks are launched from space
- ○ **B.** Because firewalls automatically allow social engineering through
- ○ **C.** Because social engineering targets people rather than technology, bypassing technical defences entirely
- ○ **D.** Because social engineering only works on Mondays

Q3 (1 mark)

How do phishing emails typically try to appear legitimate?

- ○ **A.** By being written entirely in capital letters
- ○ **B.** By copying logos, formatting, and writing styles from genuine organisations
- ○ **C.** By including the attacker's real name and address
- ○ **D.** By being sent at exactly midnight

Q4 (1 mark)

What type of information might a blagger research about their target before making contact?

- ○ **A.** Names of colleagues, job role, current projects, and organisational structure
- ○ **B.** The target's favourite colour and shoe size
- ○ **C.** The weather forecast for the next month

- ○ **D.** The history of ancient civilisations

Q5 (1 mark)

What physical actions can help prevent shouldering attacks?

- ○ **A.** Wearing brightly coloured clothing
- ○ **B.** Leaving computers unattended in public places
- ○ **C.** Speaking loudly when entering passwords
- ○ **D.** Shielding screens and keypads, using privacy screen filters, and being aware of who is nearby

Q6 (1 mark)

Why does a computer virus require a host program or file to spread?

- ○ **A.** Because viruses cannot execute on their own and need legitimate code to run them
- ○ **B.** Because viruses are too large to exist independently
- ○ **C.** Because viruses need host programs to generate electricity
- ○ **D.** Because host programs make viruses run faster

Q7 (1 mark)

Where are trojans commonly distributed from?

- ○ **A.** Official operating system updates from Microsoft or Apple
- ○ **B.** Government websites only
- ○ **C.** Physical shops selling boxed software
- ○ **D.** Unofficial download sites, pirated software, email attachments, and fake advertisements

Q8 (1 mark)

What distinguishes spyware's behaviour from a typical virus?

- ○ **A.** Spyware is designed for long-term hidden surveillance rather than replication and spreading
- ○ **B.** Spyware only affects computers in one country
- ○ **C.** Spyware makes computers run faster
- ○ **D.** Spyware can only be installed by the computer's owner

Q9 (1 mark)

What technical mechanism do pharming attacks manipulate?

- ○ **A.** The brightness settings of computer monitors
- ○ **B.** DNS (Domain Name System) settings that translate website names into IP addresses
- ○ **C.** The speed of the computer's processor

- ○ **D.** The size of the computer's hard drive

Q10 (1 mark)

Which of these represents typical default passwords that create security vulnerabilities?

- ○ **A.** 'admin', 'password', '1234', or the manufacturer's name
- ○ **B.** Long random strings like 'Kj8$mNq2!xLp9Wz'
- ○ **C.** Passwords that change automatically every hour
- ○ **D.** Passwords requiring fingerprint verification

Q11 (1 mark)

Why are removable media devices a threat even to systems not connected to the internet?

- ○ **A.** Because removable media generates harmful radiation
- ○ **B.** Because removable media changes the colour of computer screens
- ○ **C.** Because removable media only works on systems connected to the internet
- ○ **D.** Because removable media can physically carry malware across the 'air gap' that isolates offline systems

Q12 (1 mark)

What do security patches primarily fix?

- ○ **A.** Known vulnerabilities in software that could be exploited by attackers
- ○ **B.** The physical scratches on computer cases
- ○ **C.** The colour scheme of software interfaces
- ○ **D.** The weight of laptop computers

Q13 (2 marks)

An estate agent receives a phone call from someone claiming to be a solicitor handling an urgent property completion. The caller knows the property address, the buyer's name, and says the bank details for the deposit transfer need to be updated immediately due to a system change. What type of attack is this?

- ○ **A.** Pharming
- ○ **B.** Shouldering
- ○ **C.** Blagging, where the attacker has researched transaction details to create a convincing scenario for redirecting payment
- ○ **D.** A legitimate request from the actual solicitor

Q14 (2 marks)

How do the methods by which viruses and trojans first enter a system typically differ?

- ○ **A.** Viruses and trojans enter systems in exactly the same way
- ○ **B.** Viruses typically arrive hidden within infected files shared from other systems, while trojans are downloaded directly by users who believe they are getting legitimate software

- ○ **C.** Trojans enter through hardware ports while viruses enter through software
- ○ **D.** Viruses can only enter through email while trojans can only enter through websites

Q15 (2 marks)

After completing a penetration test, what should testers provide to the organisation?

- ○ **A.** Nothing-penetration testers keep their findings secret
- ○ **B.** Only a verbal summary with no written documentation
- ○ **C.** A detailed report documenting discovered vulnerabilities, their severity, evidence of exploitation, and recommendations for remediation
- ○ **D.** A list of employees to fire

Q16 (2 marks)

A family installs a smart doorbell camera but keeps the default password. Why does this create a security risk beyond just the doorbell itself?

- ○ **A.** The doorbell might ring too loudly
- ○ **B.** The camera might take blurry pictures
- ○ **C.** Attackers could access the camera feed to observe when the house is empty, monitor who visits, and potentially use the compromised device to attack other devices on the home network
- ○ **D.** The doorbell would use too much electricity

Q17 (2 marks)

A tourist in a city centre appears to be photographing architecture but is actually capturing images of people entering door codes at a nearby apartment building. What type of attack is this?

- ○ **A.** Phishing
- ○ **B.** A virus infection
- ○ **C.** Pharming
- ○ **D.** Shouldering, using a camera to observe and record private information from a distance

Q18 (2 marks)

An employee downloads a free browser extension promising to block advertisements. After installation, the employee notices their browser behaving strangely-displaying different adverts and redirecting searches to unfamiliar sites. What has most likely happened?

- ○ **A.** The extension is working exactly as advertised
- ○ **B.** The employee's internet connection has become faster
- ○ **C.** The extension is a trojan-it promised ad blocking but actually injects advertisements and redirects traffic for the attacker's profit
- ○ **D.** The computer's monitor is malfunctioning

Q19 (2 marks)

Why is using the same password across multiple online accounts dangerous?

- ○ **A.** Websites will stop working if they detect duplicate passwords

- ○ **B.** Using the same password makes computers run more slowly
- ○ **C.** If one account is compromised in a data breach, attackers can use those credentials to access all other accounts sharing the same password
- ○ **D.** Identical passwords take up more storage space on computers

Q20 (2 marks)

What is the key technical difference between how phishing and pharming direct victims to fake websites?

- ○ **A.** Phishing only works on Tuesdays while pharming works any day
- ○ **B.** Phishing requires victims to click malicious links, while pharming corrupts DNS to redirect traffic even when correct addresses are typed
- ○ **C.** Pharming requires victims to click links while phishing corrupts DNS
- ○ **D.** There is no difference-they are identical attacks

Q21 (2 marks)

How might a virus spread between computers on the same office network?

- ○ **A.** Through infected files on shared network drives that multiple users access and open
- ○ **B.** By floating through the air between computers
- ○ **C.** By traveling through the electrical power cables
- ○ **D.** Through the office telephone system

Q22 (2 marks)

Which signs might indicate a computer is infected with spyware?

- ○ **A.** The computer becomes physically heavier
- ○ **B.** The computer's screen turns permanently green
- ○ **C.** The keyboard starts typing in a different language automatically
- ○ **D.** Unexplained network activity, slower performance, browser homepage changes, and unexpected pop-ups

Q23 (2 marks)

Why does calling back on an independently obtained phone number help defeat blagging attacks?

- ○ **A.** Phone calls cannot be used for blagging attacks
- ○ **B.** Because phone companies automatically block all blagging attempts
- ○ **C.** Because attackers cannot make phone calls last more than five minutes
- ○ **D.** Because it breaks the attacker's control of the conversation-they cannot intercept a call to a number they did not provide, confirming whether the original request was genuine

Q24 (3 marks)

A dental practice has these security issues: (1) Reception staff have had no training on recognising suspicious emails; (2) The practice management software has not been updated in two years; (3) All staff share one login account for patient records. Analyse the combined security risks these issues create.

- ○ **A.** These practices improve efficiency and pose no security risks
- ○ **B.** Only the software updates matter; the other issues are minor
- ○ **C.** These create serious risks: untrained staff may fall for phishing leading to malware infection; unpatched software contains known exploitable vulnerabilities; shared logins eliminate accountability and mean one compromised password exposes all patient records-together enabling attackers multiple paths to sensitive medical data
- ○ **D.** Dental practices cannot be targets of cyber attacks

Q25 (3 marks)

An online retailer wants to test security of their systems. They have: (1) A public website where customers browse and purchase products; (2) An admin panel where staff process orders and manage inventory. What penetration testing approaches are needed and why?

- ○ **A.** No testing is needed because e-commerce sites are automatically secure
- ○ **B.** Testing the public website without credentials assesses external attack surface; testing the admin panel with staff-level credentials assesses whether employees could access or modify data beyond their authorisation-both are needed because they face different threat models
- ○ **C.** Only the admin panel needs testing because customers cannot be hackers
- ○ **D.** Testing should only check if the website loads quickly

Q26 (3 marks)

A company's accounts payable department receives an email appearing to be from a regular supplier, explaining that their bank has changed and providing new payment details for all future invoices. The email uses

the supplier's logo and mentions recent genuine orders by reference number. Analyse this attack and explain how it could be verified.

- ○ **A.** This is definitely legitimate because it mentions real order numbers
- ○ **B.** Suppliers cannot be impersonated by attackers
- ○ **C.** Bank details never need to be verified when changing
- ○ **D.** This is a supplier impersonation attack (blagging via email) using researched details for credibility; verification requires contacting the supplier using independently obtained contact details-not those in the suspicious email-to confirm whether they actually changed banks

Q27 (3 marks)

A veterinary clinic has these issues: (1) Staff use shared tablets to access pet owner records and payment information; (2) A USB drive is used to transfer X-ray images between the imaging computer and the main system; (3) The WiFi password is printed on a sign in the waiting room for client convenience. Analyse the security risks.

- ○ **A.** These practices are standard and create no security concerns
- ○ **B.** Shared tablets without individual logins eliminate accountability and risk credential theft; the USB drive could carry malware between systems; the public WiFi password allows anyone-including attackers-to access the network and potentially reach systems containing client and payment data
- ○ **C.** Only the WiFi password is a minor concern
- ○ **D.** Veterinary clinics hold no data worth protecting

Q28 (3 marks)

A football club discovers that the 'fixture scheduling software' a volunteer downloaded three months ago was actually a trojan. Investigation reveals it contained spyware that has been collecting data from the membership database including names, addresses, dates of birth, emergency contacts, and direct debit details of 800 members. Analyse the impacts.

- ○ **A.** The only impact is that match schedules might be incorrect
- ○ **B.** Sports club members have no expectation of data protection
- ○ **C.** This is minor because sports clubs do not hold important data
- ○ **D.** Serious impacts include: 800 members face identity theft risk from personal data exposure and potential financial fraud from direct debit details; the club must notify affected members and the ICO; members will lose trust and may leave; the club faces potential regulatory fines and legal action; the volunteer's well-meaning action has caused significant harm

Q29 (3 marks)

A university notices increased reports of students receiving emails claiming their student accounts will be suspended unless they verify their login details via a provided link. The emails use university branding and mention the current academic term. Some students have clicked the links and entered their credentials. Analyse this attack and its potential consequences.

- ○ **A.** This is a phishing campaign using university branding and urgency to harvest student credentials; consequences include: compromised accounts may be used to access grades, personal data, and financial information; attackers could send further phishing from trusted student accounts; university systems may be accessed through stolen credentials; and students whose credentials were captured need immediate password resets
- ○ **B.** This is helpful IT communication and students should follow the instructions
- ○ **C.** Only international students would be affected by this attack

○ **D.** Universities cannot be targets of phishing attacks

Q30 (3 marks)

An engineering company discovers that for six months, confidential product design files have been copied and transmitted to servers in another country. Investigation traces the breach to an email attachment opened by a design engineer-a fake job application that installed malware. Trace the complete attack chain and analyse the impacts.

○ **A.** This was just an innocent job application with no malicious intent

○ **B.** The attack chain was: phishing email (fake job application) delivered a trojan (disguised as a document) which installed spyware (capturing and transmitting design files); impacts include loss of competitive advantage if designs reach competitors, potential contract breaches if client designs were stolen, regulatory concerns if export-controlled technology was transmitted, and damage to client relationships and company reputation

○ **C.** Only the design engineer who opened the email is affected

○ **D.** Manufacturing companies do not store valuable data

Total: **55** marks • Time: **30 min**

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